

4 Perceptron

1. (a) Since $f = w^\top x$ and writing $w = \sum \alpha_i x_i$ implies that $f = \sum \alpha_i \langle x_i, x \rangle$

(b) Since update for the unkernelized update is $w \leftarrow w + y_i x_i$ with $w = \sum \alpha_i x_i$ on the j th iteration the update becomes $\sum_{i=1}^{j-1} \alpha_i x_i + y_j x_j$. This implies that $\alpha \leftarrow \alpha + y_j$ or in the code below if $\alpha'_i y_i := \alpha_i$ then α' is the vector of mismatches that occur for point x_i as the algorithm evolves.

(c) For the kernelized version the calculation of $f(x)$ becomes

$$f = \sum \alpha_i \langle x_i, x \rangle \mapsto \sum \alpha_i K(x_i, x)$$

instead of $f = w^\top x$.

2. The code and subsequent plots are below.